# Overview

The methodology used for this analysis is designed to mimic the methodology used by the Bureau of Labor Statistics to conduct Occupational Projections with the exception of the models used. By changing the models used from statistical and econometric models to machine learning models which are more focused on accuracy, this analysis can be used to identify the model which would be an effective benchmark to further ensure the accuracy of the BLS method's projections. This version of the projections methodology covers the labor force inputs, aggregate economy inputs, final demand proxy, and industry employment inputs used in the BLS methodology. The limits of this analysis lead to a divergence from the BLS methodology in that this analysis did not use paid multipliers from IMPLAN or RIMS II. The further divergence of the type of models used is implemented in the Occupational Employment block. By training on several years approaching 2024, and testing on 2024 data we will be able to evaluate the prediction of the models against the actual employment numbers since these have been published at the time of this analysis.

# 2. Data Acquisition

## 2.1 Scope of Data

The data collected for this analysis is statewide data on Florida ranging from 2016 to 2024. All data being collected is annual units on detailed occupations. The target variable in this analysis is the occupational employment by industry for each year. This means we are trying to predict the employment of a certain occupation within a certain industry. 2015 data is excluded because the BLS OEWS research estimates files do not have detailed occupation groups for that year. The training data set is 2016-2023 and the test year is 2024.

## 2.2 BLS Methodology

### 2.2.1 Labor Force Inputs

The features used in this analysis from the Labor Force Inputs block of the BLS methodology include the Annual Averages of Civilian Noninstitutional Population 16 years and older (CNP) (Series ID: LAUST120000000000009), Labor Force Participation Rate (LFPR) (Series ID: LAUST120000000006), and Total Labor Force Level (Series ID: LAUST120000000008). These inputs are necessary to get the context of the labor force and a clear picture of the supply for the labor force as well as an idea of how likely that supply is to grow. The features come from the BLS' own Local Area Unemployment Statistics survey (LAUS) using period M13.

### 2.2.2 Aggregate Economy

The features used in this analysis from the Aggregate Economy block of the BLS methodology include the Real and Nominal Gross State Products, State Personal Consumption Expenditures, Government GDP by Industry, Construction GDP as an Investment Proxy, and the GDP Implicit Price Deflator. These inputs are a collection of economic indicators as well as industry-level detail indicators that are used in the estimation process for industry-level employment projections. All of the features included in this block come from the Bureau of Economic Analysis (BEA).

These features were retrieved using the bea.R package to access the BEA API. The features are retrieved from the Regional dataset: SAGDP2 for nominal GDP by industry, SAGDP9 for real GDP by industry, SAPCE1 for PCE, SAIMC4 for total employment, and Table T10109 from the NIPA dataset for the GDP deflator.

### 2.2.3 Final Demand

BLS uses multipliers purchased from IMPLAN or BEA (RIMS II), however because of the limitations on this analysis, the feature used to bridge the gap as a proxy for demand across NAICS industries is the previously mentioned real and nominal GDP by industry collected from the BEA.

### 2.2.4 Industry Employment

The features used in this analysis from the Industry Employment block of the BLS methodology include employment and wages by industry, total employment including proprietors, and the national average weekly hours worked by industry. These features are included in this block to act as a benchmark for the sum of the predicted employment by occupation and industry. Industry level employment and wages come from the BLS' Quarterly Census of Employment and Wages (QCEW), the total employment including proprietors is coming from the BEA, and the national average weekly hours worked by industry is coming from the BLS' Current Employment Statistics (CES).

All data in this section is retrieved through the BLS API. QCEW data is annual averages filtered to the private sector and 4-digit NAICS codes. The CES weekly hours are retrieved at the supersector level.

# 3. Models Evaluated

# 3.1 Models

Each model uses the same features. The goal is to compare their results against each other and against BLS' projections when compared to the actual employment numbers for 2024.

### 3.1.1 OLS

This is used as a baseline model to mimic the current BLS approach of using traditional regression methods. The results of this model will be compared to more complex models.

### 3.1.2 Ridge and Lasso Regression

These models share a penalty logic which decreases the coefficients to account for correlated predictors. Lasso regression can decrease coefficient down to zero to remove less useful predictors.

### 3.1.3 XGBoost

This model captures nonlinear relationships. It sequentially builds decision trees which correct the errors of the previous trees. The hyperparameters for this model are tuned using cross-validation using a latin hypercube grid of 30 combinations.

# 3.2 Evaluation Metrics

The metrics are used to evaluate the held out 2024 data set and then also applied to the BLS projections.

### 3.2.1 RMSE

This is the primary metric used to select the best model. It applies a heavy penalty to large errors.

### 3.2.2 MAE

This is the average of the differences between the predicted and actual values with number of jobs as the unit.

### 3.2.3 $R^2$

This is the proportion of the variance explained by the model.

# 4. Tools

## 4.1 Data Retrieval

The data for this analysis was retrieved through manual downloads from the Bureau of Labor Statistics, Bureau of Economic Analysis, and U.S. Census Bureau websites.

## 4.2 Data Processing

The R programming language in RStudio is being used for this analysis. For data cleaning and reshaping to merge data files, I am using the tidyverse package for R. The httr and jsonlite packages are being used to pull data through APIs. The readxl package is being used to pull the OEWS Research Estimates files from a local drive.

## 4.3 Modeling

For the OLS model, the function being used is the base R function “lm”. The function being used for both the Ridge and Lasso regression models is “glmnet” called via tidymodels. Finally, the function used for the XGBoost model is “xgboost” also called via tidymodels.

## 4.4 Cross-Validation

The data is split by group_vfold_cv grouped by Year. This method was chosen because the data is structured by occupation and year. Having the same occupation in adjacent years on both sides of the split creates leakage through serial correlation. Holding out entire years at a time ensures the model is evaluated on a year not already included in the model.

## 4.5 Evaluation Metrics

The evaluation metrics included in this analysis are RMSE, MAE, $R^2$ via tidymodels metric_set. These evaluation metrics are applied to the held-out 2024 test data set.

## 4.6 Parallelization

The future package is used for parallel processing during hyperparameter tuning to reduce computation time.